INCH-POUND

A-A-55211

4 June 1993

COMMERCIAL ITEM DESCRIPTION

YARN, WOOL

The General Services Administration has authorized the use of this Commercial Item Description in preference to MIL-Y-16654.

Abstract This Commercial Item Description covers worsted yarn intended for use in hand weaving, textile work in occupational therapy, and lubricating purposes.

Classification. The yarn shall be of the following types:

Type I - White or dyed
Type II - Natural

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FSSD, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8310

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Salient Characteristics. The wool top shall be fleece wool, pulled wool, or any combination thereof not lower in grade than 56's U.S.Standard. The use of noils or laps is prohibited. The finished yarn shall contain not less than 95 percent wool based on the dry weight of the desized specimen. The yarn shall be spun on the worsted system from combed top on either the Bradford, French, or American system. The yarn shall be not finer than 8's 4 ply or coarser than 7's 4 ply. The twist shall be soft with 5 turns per inch (±5 percent) in the single yarn and 2 turns per inch (±5 percent) in the ply yarn. The color of the Type I yarn shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation of the standard sample under incandescent lamplight at 2800° Kelvin. When white is specified, the yarn shall be bleached using hydrogen peroxide. Unless otherwise specified, the white yarn shall not be treated with an optical bleach. The Type II yarn shall be in the natural condition, undyed, unbleached and not subjected to any chemical processes.

Table I Physical Requirements and Test Methods

Characteristic	Requirement	Test Method	
Yarn count	7/4 - 8/4		1\
Turns per inch			
singles	5		1\
ply	2		1\
Hydrogen Peroxide bleach			1\
Optical bleach		FED-STD 191-5660	2\
Colorfastness to			
light (Type I only)	good	AATCC 16A	3\ 4\
laundering	good	AATCC 61 Test 1A	3\ 4\
Wool content	at least 95 percent	FED STD 191-2101	

1\ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2\ In the event that the use of an optical bleach is specified or permitted, the optical bleach shall be so selected that the finished yarn shall show no discoloration from light after exposure for 20 standard fading hours followed by 48 hours in the dark.

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3\ The dyed and finished yarn shall show fastness to laundering and to light equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished yarn shall show "good" fastness to laundering and to light.

4\ Use 2 to 6 grams of the applicable material held together to form a unit for testing unless otherwise specified in the procurement document. The laundering cycle of the launderometer or similar machine shall be 30 minutes and the temperature of the container and its contents shall be maintained at 100° ±4° F.

Label/marking. Yarn manufactured under this Commercial Item Description shall be labeled in accordance with the Wool Products Labeling Act of 1939.

Workmanship. The finished yarn shall conform to the quality and grade of product established by this Commercial Item Description. The occurrence of defects shall not exceed the applicable acceptable quality levels. After completion, the finished yarn shall be thoroughly cleaned, and all loose fiber, lint, and foreign matter removed.

Regulatory Requirements. The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Quality Assurance

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own specifications, standards and quality assurance practices. The government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for, under the provisions of the contract. Reliance on contractor QA systems shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract.

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Put-up. Unless otherwise specified, the yarn for Types I and II shall be put-up in 4.0 ounce skeins (±10 percent). The average weight per skein, on a lot basis, shall be not less than 4.0 ounces. A lot shall be unacceptable if the average weight is less than the specified weight. The tolerance of plus or minus 10 percent allowed per individual skein shall not apply to the average weight. A defect with regard to weight shall be considered to exist if the net weight of any one skein is not within the specified range. The inspection level shall be S-3. The Acceptable Quality Level (AQL) shall be 2.7 defects per 100 units. For lots consisting of 500 or fewer units, the sample size shall be 10 and the acceptance number 1. The lot size shall be the number of units in the inspection lot.

Examination. Yarn skeins shall be inspected in accordance with "Sampling Procedures and Tables for Inspection by Attributes", ANSI/ASQC Z1.4, published by the American Society for Quality Control.

In-process inspection. Inspections shall be made at any point or during any phase of the manufacturing process to assure that no noils or laps are used and that the dyeing requirements have been adhered to. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance.

Visual Examination. Defects found during this examination shall be classified in accordance with the defects listed in Table II and regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the most serious defect shall be counted. The sample unit shall be one skein. The Acceptable Quality Level (AQL) shall be 1.5 major defects and 6.5 total (major and minor combined) defects per 100 units. The sample size shall be in accordance with inspection level S-1. The lot size shall be expressed in units of one skein.

Table II Classification of defects

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Cut, any	X	
	Chafed or damaged affecting serviceability	X	
	Put-up other than specified	X	
	Improperly or not uniformly wound resulting in entangling during unwinding or otherwise affecting free unhampered unwinding of the skein.	X	

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Color	Other than specified	X	
	Off shade, i.e. not within established tolerance		X
Cleanliness	Spot or stain 1\		X
	Lumps 1\		X
	Objectionable odor		X
Identification (fiber)	Not labeled in accordance with Wool Products Labeling Act of 1939		X

1\ Clearly visible at normal inspection distance (3 feet)

Wool. The wool shall be visually examined for grade in top form by comparison with the U.S.Standard Methods of Test for Grades of Wool Top. In the event of a dispute, the wool grade shall be determined by the width method (wedge). The sample unit shall be one yard of top. The sample size shall be as shown in Table III. The lot size shall be expressed in units of one pound. The result from each sample unit shall be reported separately. The lot shall be unacceptable if a single sample unit fails to meet the required grade.

Table III

Lot size (pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Tests. The methods of testing specified, wherever applicable, and as listed in Table I shall be followed. The physical and chemical values specified apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be two skeins. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The lot size shall be expressed in units of one skein. All test reports shall contain the individual values utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

Lot size (skeins)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Packaging Examination. An examination shall be made to determine that the packaging, packing, and marking complies with all listed requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the Acceptable Quality Level shall be 1.5 defects per 100 units in accordance with ANSI/ASQC Z1.4.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling or improper taping.
	Bulged or distorted container.
	Open and non-continuous heat sealed seams and closure of polyethylene bag. Omission of vent hole in polyethylene bag. Incorrectly fabricated bag.
Content	Number of skeins per package is more or less than required.
Preservation	Preservation improper, improperly applied or missing. 1\

1\ For this defect one package shall be examined from each shipping container in the sample.

Preservation, Packing, and Marking. The preservation, packing, and marking shall be as specified in the contract or order.

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Source of Government Documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

Methods of Test for Grades of Wool Top

(Applications for copies should be addressed to the United States Department of Agriculture, Agriculture Marketing Service, Washington, D.C. 20402.)

Rules and Regulations Under the Wool Products Labeling Act of 1939

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

Sources of Nongovernment Documents.

ANSI/ASQC Z1.4 - Sampling Procedures and Tables For Inspection By Attributes

(Applications for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018-3308.)

AATCC tests

(Applications for copies should be addressed to American Association of Textile Chemists and Colorists (AATCC), P.O.Box 12215, Triangle Park, NC 27709-2215.)

Custodians:
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Project No. 8310-0195

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-55211	93/6/4

3. DOCUMENT TITLE

YARN, WOOL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Personnel Support Center ATTN: DPSC-FSSD (12-3-D) - DLA-CT	(1) Commercial 215-737-8105	(2) AUTOVON - DSN 444-8105
c. ADDRESS (Include Zip Code) 2800 South 20th Street P. O. Box 8419 Philadelphia, Pennsylvania 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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